ANUSANDHAN NATIONAL RESEARCH FOUNDATION

PM ECRG – YOUNG RESEARCHER PROPOSAL

BIOSENSORS FOR EARLY DISEASE DETECTION

Development of a Sensitive, Low-Cost and Portable Electrochemical Biosensor Platform

Principal Investigator	Tanmay Shailendra Patil
Project Duration	3 Years
Research Area	Biosensors • Nanotechnology • Biomedical Diagnostics
Proposed Scheme	ANRF – Prime Minister Early Career Research Grant

A modern biomedical diagnostics concept focused on early, accessible and rapid disease detection.

1. Project Summary

This project proposes the development of a multiplexed electrochemical biosensor platform for early detection of clinically relevant disease biomarkers. The system will combine nanostructured sensing materials with selective biorecognition elements to achieve rapid, sensitive and low-cost measurements. The intended outcome is a portable proof-of-concept platform suitable for future point-of-care diagnostic development.

2. Aim

To develop a sensitive, low-cost and portable electrochemical biosensor platform for early disease detection.

3. Specific Objectives

01	Design and fabricate nanostructured electrochemical sensing electrodes.
02	Develop selective biomarker-recognition strategies using suitable bioreceptors.
03	Optimize sensor sensitivity, selectivity, response time and reproducibility.
04	Validate the prototype using laboratory-prepared and simulated biological samples.

4. Novelty & Innovation

The proposed work integrates nanomaterial-enhanced electrochemical transduction with multiplexed biomarker detection in a compact platform. Emphasis will be placed on low-cost materials, simple fabrication, miniaturization and analytical performance suitable for future point-of-care translation.

5. Methodology

Sensor Design	Selection of electrode architecture, nanomaterials and surface chemistry.
Functionalization	Immobilization of antibodies, aptamers or other suitable recognition elements.
Characterization	Electrochemical and physicochemical characterization of sensing interfaces.
Optimization	Evaluation of sensitivity, selectivity, linearity, response time and stability.
Validation	Testing with spiked/simulated biological samples and comparison with reference measurements where feasible.

Concept flow: Biomarker → Bioreceptor → Nanostructured electrode → Electrochemical signal → Quantitative detection

6. Three-Year Work Plan

Phase	Year 1	Year 2	Year 3
Design & fabrication	●	●	
Surface functionalization	●	●	
Optimization & analytical testing		●	●
Sample validation			●
Publication / IP / translation		●	●

7. Expected Outcomes

• A laboratory-scale electrochemical biosensor prototype.

• Optimized sensing interface with improved analytical performance.

• Proof-of-concept multiplexed detection strategy.

• Research publications and potential intellectual property.

• Foundation for future portable point-of-care diagnostic development.

8. Illustrative Budget

Component	Amount
Equipment	■17.0 lakh
Research Personnel	■10.8 lakh
Consumables	■17.0 lakh
Travel	■2.5 lakh
Contingency	■2.2 lakh
Subtotal	■49.5 lakh
Overheads	As per applicable ANRF norms

9. Ethical & Translational Considerations

The initial work will emphasize laboratory and simulated biological samples. Any future study involving human participants or clinical specimens will require appropriate institutional ethics approval, informed consent and compliance with applicable biosafety and research regulations. Translation will focus on affordability, portability, reproducibility and usability.

10. Conclusion

The proposed research aims to establish a practical biosensor platform for early disease biomarker detection by combining nanotechnology and electrochemical sensing. Successful completion will provide a strong foundation for

further validation, publications, intellectual property and future point-of-care diagnostic development.

Institution: _____

Department: _____

Date: _____ **Signature:** _____